

EMBOLISM

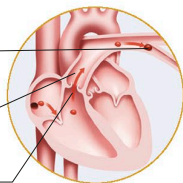
Most emboli are fragments of a blood clot (thrombus), or even a whole clot, that has detached from its original site and travelled in the bloodstream to lodge in a blood vessel. An embolus may also be made of fatty material from an atheromatous plaque (see p.157) in an arterial wall, crystals of cholesterol, fatty bone marrow that has entered the circulation following a bone fracture, or an air bubble or amniotic fluid. In a pulmonary embolism, a clot originating elsewhere in the body travels in the veins to the lungs. Clots that form in the heart or arteries can block the blood circulation anywhere in the body. An embolus is most likely to block a

blood vessel where it narrows or branches, depriving the tissues of oxygen beyond the site of the blockage. The symptoms of an embolism depend on the site affected.

Embolus
travelling to lung

Pulmonary artery

Path of embolus



PULMONARY EMBOLISM

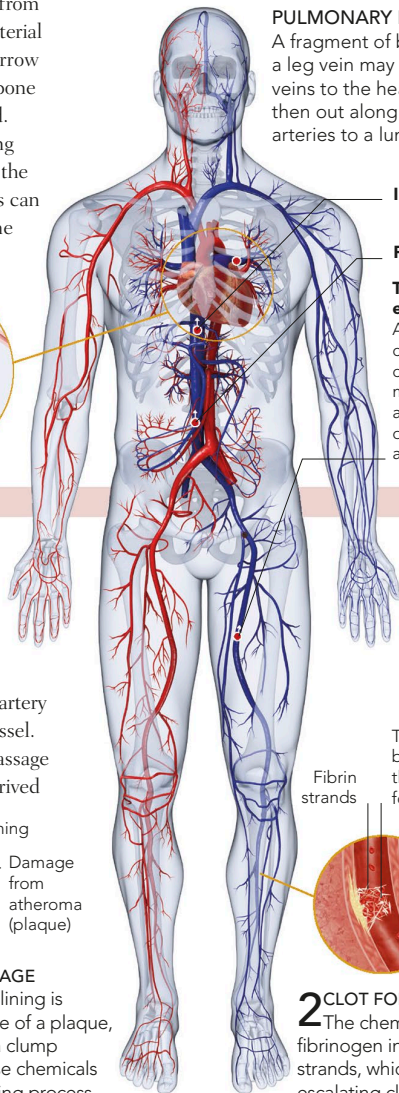
A fragment of blood clot from a leg vein may travel through the veins to the heart's right side, then out along the pulmonary arteries to a lung.

Inferior vena cava

Path of embolus

Thrombotic embolus

A fragment (embolus) composed of blood clot (thrombotic) material may arise anywhere, but veins of the legs and pelvis are common sites



THROMBOSIS

Thrombosis is the blockage of a blood vessel by a blood clot. It is most likely to occur where normal blood flow is disrupted, which may be due to plaques of fatty atheromatous tissue in the walls of an artery (see p.157) or inflammation of the blood vessel. The clot eventually narrows or blocks the passage for blood so that the tissues beyond are deprived of oxygen and nutrients.

Platelet

Lining

Damage
from
atheroma
(plaque)

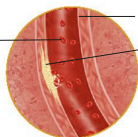
THROMBUS FORMATION

Thrombosis can occur in arteries and veins, but commonly happens at a site of atherosclerosis in an artery wall, which disrupts the normal blood flow.

1

INTERNAL DAMAGE

When an artery lining is damaged by rupture of a plaque, platelets in the area clump together and release chemicals that begin the clotting process.



Fibrin
strands

Thrombus
blocking artery;
thrombi can also
form in veins

2

CLOT FORMATION

The chemicals help convert fibrinogen into insoluble fibrin strands, which trap blood cells, escalating clot formation.

